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SIMATS ENGINEERING

CO3 - ASSESSMENT TOOL 3

Comparative Analysis

COURSE NAME: DEEP LEARNING
SLOT :D

COURSE CODE: MLA0405

COURSE OUTCOME COVERED

CO3: Students will be able to understand, analyze and apply CNN concepts including layers, filters, parameter sharing, regularization, AlexNet and ResNet for real-world image processing applications. (BTL-6)

CO Weightage: 25%

Duration: 90 minutes

Assessment Tool Weightage: 40%

Marks: 25

Code of Conduct:

I Ch. Venkata Lakshmi Hymavathi (192425397) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Name of the student:

Ch. Hymavathi

Ch. Venkata Lakshmi Hymavathi

Criteria	Selection of Studies/Parameters (2)	Comparison & Differentiation (2.5)	Critical Evaluation (2.5)	Synthesis & Insight Generation (2)	Clarity & Presentation (1)	Total (10)
Weightage	20 %	25%	25%	20 %	10 %	100%
Q1						
Q2						
Q3						
Q4						
Q5						
Total						

Signature of the Faculty:

Total Marks :

QUESTIONS: 05**Total Marks:25****Q1. CNN vs Fully Connected Neural Network**

A medical imaging organization wants to classify X-ray images using deep learning. Compare a **CNN with a traditional fully connected neural network** in terms of architectural design, motivation, spatial feature extraction, number of parameters, computational complexity, and ability to handle image data. Analyze the roles of convolutional layers, filters, and pooling layers, and justify why CNNs are more suitable for image classification applications.

Answer:

Aspect	CNN	Fully Connected Neural Network
Architecture	Uses convolution, pooling and fully connected layers	Mostly consists of fully connected layers
Motivation	Designed specifically for grid-like data such as images	Designed for general numerical/vector data
Spatial features	Preserves spatial relationships between pixels	Loses spatial arrangement after flattening
Parameters	Fewer because of parameter sharing	Very large number of parameters
Computation	More efficient for images	Computationally expensive for high-resolution images
Image handling	Excellent	Less suitable
Feature learning	Automatically learns edges, textures and shapes	Learns features but without explicitly considering locality

Role of CNN layers

1. **Convolutional layers:** Apply filters to the image and detect local patterns such as edges, corners and textures.
2. **Filters:** Small matrices that slide across the image to extract specific features.
3. **Pooling layers:** Reduce the spatial size of feature maps, decreasing computation while retaining important information.
4. **Fully connected layers:** Use the extracted features to perform final classification.

Justification

CNNs are more suitable for X-ray image classification because they **preserve spatial information, use fewer parameters, automatically learn visual features, and efficiently process large images**. Therefore, CNNs generally provide better performance for medical image classification than traditional fully connected networks.

Q2. Different CNN Layers and Filters

A manufacturing company is developing a CNN-based system to detect defects in industrial products. Compare the functions of **convolutional layers, pooling layers, activation layers, and fully connected layers** in terms of feature extraction, spatial information, computational requirements, and classification. Further compare **early-layer filters and deeper-layer filters** based on the type of features they learn, and justify how these layers collectively improve defect detection.

Answer:

Layer	Main Function	Spatial Information	Computation
Convolution	Extracts features	Preserves spatial structure	Moderate
Pooling	Reduces feature-map size	Retains important information	Low
Activation	Adds non-linearity	Maintains feature-map structure	Low
Fully Connected	Final classification	Converts features into class predictions	Higher

1. Convolutional Layer

Convolutional layers apply filters to the input image and identify useful patterns such as edges, curves, textures and defect regions.

2. Pooling Layer

Pooling reduces the height and width of feature maps. **Max pooling**, for example, selects the maximum value from a small region. This reduces computation and makes the network less sensitive to small changes in position.

3. Activation Layer

Functions such as **ReLU** introduce non-linearity. This allows the CNN to learn complex patterns instead of only simple linear relationships.

4. Fully Connected Layer

The extracted features are converted into class predictions such as **defective** or **non-defective**.

Early vs Deep Filters

- **Early-layer filters:** Learn simple features such as edges, lines, corners and basic textures.
- **Middle-layer filters:** Combine simple features into patterns such as shapes and surface structures.
- **Deep-layer filters:** Learn complex and high-level features representing specific defects or product characteristics.

Justification

The layers work hierarchically: **convolution extracts features** → **activation learns complex relationships** → **pooling reduces unnecessary information** → **fully connected layers classify the product**. Therefore, the complete CNN can effectively identify both simple and complex industrial defects.

Q3. Parameter Sharing vs Independent Weights

A smart surveillance system needs to process thousands of high-resolution images efficiently. Compare **parameter sharing in CNNs with independent weights in fully connected networks** in terms of the number of parameters, memory requirements, computational complexity, feature detection, and translation-related invariance. Analyze how the reuse of convolutional filter weights improves learning efficiency and justify its importance for large-scale image-processing applications.

Answer:

Aspect	CNN Parameter Sharing	Fully Connected Independent Weights
Parameters	Much fewer	Very large
Memory	Low	High
Computation	Efficient	Expensive for large images
Feature detection	Detects the same feature at different locations	Learns location-specific relationships
Spatial structure	Preserved	Largely lost after flattening
Translation handling	More robust to feature movement	Less robust
Learning efficiency	High	Lower for image data

Parameter Sharing in CNN

In a CNN, the same filter weights are reused across different locations of an image. For example, one edge-detection filter can detect that edge whether it appears on the left, center, or right side.

Instead of learning separate weights for every pixel connection, CNNs learn a small set of filter weights and reuse them.

Advantages

1. Fewer parameters: Reduces the size of the model.
2. Lower memory requirement: Fewer weights need to be stored.
3. Faster learning: The same filter learns from many locations.
4. Better feature detection: Important features can be detected anywhere in the image.
5. Translation robustness: Small movements of an object do not completely change the learned representation.

Justification

For thousands of high-resolution surveillance images, parameter sharing is extremely important because independent weights would require enormous memory and computation. CNNs therefore provide efficient, scalable and effective image processing.

4. AlexNet vs ResNet

A computer vision company is developing an image classification system and is considering **AlexNet and ResNet** as possible architectures. Compare the two architectures in terms of network depth, layer organization, convolutional filters, parameter requirements, feature extraction capability, training difficulty, vanishing-gradient problem, computational cost, and classification performance. Recommend the more suitable architecture for a complex image recognition application and justify your choice.

Answer:

Aspect	AlexNet	ResNet
Depth	8 learned layers	Can have 18, 34, 50, 101, 152+ layers
Organization	Conventional sequential CNN	Uses residual/skip connections
Filters	Mainly 11×11 , 5×5 and 3×3 filters	Mainly smaller 3×3 filters in many versions
Parameters	About 60 million	Depends on version; ResNet-50 has about 25.6 million

Feature extraction	Good for basic/medium complexity	Excellent for complex hierarchical features
Training difficulty	Easier due to shallower depth	Skip connections make deep training easier
Vanishing gradient	Less severe because network is shallower	Skip connections greatly reduce the problem
Computational cost	Lower for the original model	Higher for deeper versions
Performance	Historically important, but older	Generally better for modern complex recognition

AlexNet

AlexNet is a relatively shallow CNN that uses convolutional layers, ReLU activation, pooling and fully connected layers. It was highly influential in demonstrating the effectiveness of deep CNNs for image classification.

ResNet

ResNet introduces **residual/skip connections**. Instead of forcing every layer to learn a complete transformation, the network can learn a residual function and pass information directly through shortcut connections.

This makes it possible to train very deep networks while reducing the **vanishing-gradient and degradation problems**.

Recommendation

For a complex image-recognition application, **ResNet is the better choice**. Its deeper architecture can learn more complex features, while residual connections make deep training more stable. Although it may require more computation, its improved feature extraction and classification performance make it more suitable for complex applications.

Q5. Regularized CNN vs Non-Regularized CNN

An agricultural organization is developing a CNN to classify plant diseases from leaf images. The initial model achieves very high training accuracy but performs poorly on unseen images. Compare a **CNN without regularization** and a **CNN using dropout, data augmentation, and batch normalization** in terms of overfitting, generalization, training stability, computational requirements, and model performance. Analyze which regularization techniques should be applied and justify the most appropriate approach for reliable real-world disease classification.

Answer:

Aspect	Without Regularization	With Regularization
Overfitting	High	Reduced
Training accuracy	Often very high	Usually high
Unseen-image performance	Poor	Better
Generalization	Low	High
Training stability	May be unstable	Batch normalization improves stability
Computation	Lower	Slightly higher
Reliability	Lower	Higher

1. CNN Without Regularization

The model may memorize the training leaf images. This results in high training accuracy but poor validation/test accuracy, which is a typical sign of overfitting.

2. Dropout

Dropout randomly disables some neurons during training. This prevents the network from depending too heavily on particular neurons and helps improve generalization.

3. Data Augmentation

Data augmentation creates varied training examples by applying transformations such as:

- Rotation
- Flipping
- Cropping
- Small shifts
- Brightness changes

This is particularly useful for plant-disease images because leaves may appear at different angles, sizes and lighting conditions.

4. Batch Normalization

Batch normalization normalizes intermediate activations during training. It can make training more stable and often allows the network to converge faster.

Recommended Approach

For reliable plant-disease classification, the CNN should use data augmentation + batch normalization + suitable dropout.

Data augmentation is especially important because it increases the diversity of training images. Dropout helps prevent overfitting, while batch normalization improves training stability.

Conclusion

A regularized CNN is more suitable for real-world agricultural applications because it generalizes better to unseen leaf images and reduces overfitting. Therefore, the organization should prefer a regularized CNN rather than relying on a non-regularized model.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Selection of Studies/Parameters	20%	Selects highly relevant and diverse studies with well-defined comparison parameters	Selects relevant studies with minor gaps in parameter definition	Limited or partially relevant selection	Poor or inappropriate selection of studies
Comparison & Differentiation	25%	Clearly compares multiple aspects (methods, results, limitations) with strong differentiation	Good comparison with minor gaps	Basic comparison with limited depth	No clear comparison; mostly descriptive
Critical Evaluation	25%	Critically evaluates strengths, weaknesses, and implications with strong justification	Provides evaluation with some justification	Limited evaluation; lacks depth	No critical evaluation provided
Synthesis & Insight Generation	20%	Generates meaningful insights and conclusions from comparisons	Some insights derived from comparison	Limited or unclear insights	No insights generated
Clarity	10%	Information is clearly presented (tables/figures) with logical structure	Generally clear with minor issues	Presentation lacks clarity or structure	Poor presentation and difficult to understand